Key features of sine, cosine, and tangent graphs

Core Concepts: Key features of sine, cosine, and tangent graphs

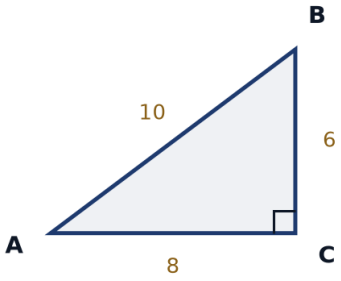
1 For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(2)$.

a. 11

b. 13

c. 4

d. 15



2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(3)$.

a. 13

b. 15

c. 6

d. 17

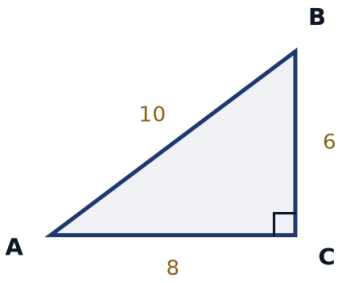
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(4)$.

a. 15

b. 17

c. 8

d. 19

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8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

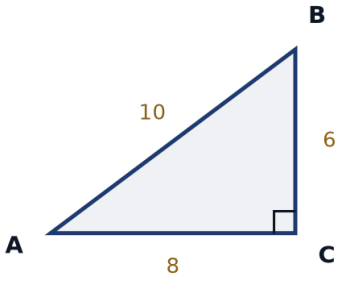
9 A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5



Exam-Style Problem Solving & Modeling

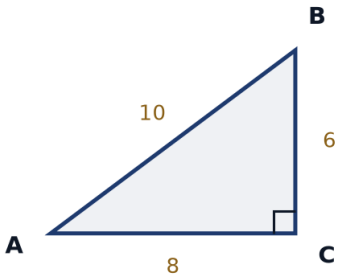
10 For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(5)$.

a. 17

b. 19

c. 10

d. 21



11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

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12 A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

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